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Genome-wide identification and expression analysis of the *SPL* transcription factor family and its expression characteristics in *Elymus sibiricus*

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Abstract

Background *Elymus sibiricus* is widely utilized for establishing of high-yield artificial grasslands due to its remarkable productivity and strong resistance to environmental stresses, making it an excellent forage species. *SPL* transcription factors play a pivotal role in regulating plant growth, development, and responses to abiotic stress. Although the *SPL* gene family has been identified in many plant species, its presence and function in *E. sibiricus* remain largely unexplored.

Result This study presents a comprehensive genome-wide identification and analysis of the *SPL* gene family in *E. sibiricus*. A total of 37 *EsSPL* genes were successfully identified. Their chromosomal distribution, gene structure, conserved motifs, cis-acting regulatory elements, and evolutionary relationships were analyzed. Protein-protein interaction network analysis predicted that SOC1 and TOE2 are the primary interacting proteins. Most *EsSPL* genes exhibited high expression levels in seedling tissues. Additionally, analysis of abiotic stress responses revealed that the expression of multiple *EsSPL* genes was altered under salt, drought, abscisic acid (ABA), and gibberellin (GA) treatments.

Conclusion Through sequence homology analyses, 37 *SPL* genes were identified in *E. sibiricus*. Most *SPL* family members exhibited high expression levels in seedling tissues, with *EsSPL2* specifically upregulated under four distinct abiotic stress conditions. These findings provide a foundation for understanding the genetic evolution and biological functions of the *SPL* gene family in *E. sibiricus*, offering valuable insights for future research and breeding efforts.

Keywords *E. sibiricus*, *SPL* gene family, Evolutionary relationships, Gene expression, Abiotic stress

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Introduction

The *SPL* (SQUAMOSA PROMOTER-BINDING PROTEIN-LIKE) gene family is a group of plant-specific transcription factors are evolutionarily conserved across embryophytes [1]. *SPL* genes encode transcription factors characterized by a conserved SBP-box domain, which facilitated DNA binding [2, 3]. The number and functional diversity of *SPL* family members vary across plant species [4]. For instance, *Arabidopsis thaliana* possessed 17 *SPL* genes [5]. Despite differences in their primary distinct protein primary structures, all *SPL* proteins share a highly conserved SBP domain, approximately 80-amino-acid in length [6]. In addition, *SPL* transcription factors play crucial roles in diverse plant growth and developmental processes including photoperiod [7, 8], gibberellin signalling pathways [9], regulation of meristem-specific genes [10, 11], and floral organ genes [12, 13], responses to heat and drought stress [14–16], leaf morphogenesis [17, 18], and the development of lateral, primary roots and adventitious roots [19].

Current research on *SPL* genes function predominantly focuses on model plants such as *A. thaliana* and *Oryza sativa*. In *A. thaliana*, miR156 and *SPL* genes play opposing roles in regulating flowering time: miR156 overexpression delays flowering, whereas *SPL* genes promote it. Specifically, *AtSPL3*, *AtSPL4*, and *AtSPL5* play critical roles in both flowering time and nutritional phase transition process. Increased expression of *AtSPL3* and *AtSPL9* accelerates flowering, while miR156 overexpression suppresses *SPL* activity, leading to delayed flowering [20, 21]. Furthermore, overexpressing *AtSPL10* promotes flowering, whereas *spl1/spl2/spl3* triple mutants exhibit delayed flowering compared to the wild type. Further studies indicate that *AtSPL10* regulates downstream targets such as *MED25*, which in turn influences flowering-related genes like *FRUITFULL* (*FUL*) and *LEAFY* (*LFY*) [22]. In rice, *OsSPL3* regulates crown root development [23]. While *OsSPL13* (*GLW7*) positively affects glume cell size, thereby enhancing grain length and yield [24]. *OsSPL14* (*IPA1*) (Ideal Plant Architecture 1), a key regulator of plant architecture, not only improves yield but also enhances disease resistance in rice [25, 26]. *SPL* genes regulate various aspects of plant architecture, including inflorescences, pedicel length, floral organ size and early flower development. In *Antirrhinum majus* (snapdragon), SBP1 and SBP2 interact with the promoter of *SQUA*, regulating early flower development by influencing *SQUA* expression [27]. Moreover, in *V. pseudoreticulata*, *VpSBP16* enhances salt and drought tolerance by modulating SOS and ROS signaling pathways when overexpressed in *A. thaliana*. This improvement is observed across multiple growth stages, including seed germination, seedling growth, and mature plants [28]. In *Malus xiaojinensis*, *MxSPL26* interacts with *MxTIFY9*, which

is induced by exogenous hormone IBA, to jointly suppress the homeobox-leucine zipper protein (*MxHB13*) expression. This interaction regulates *MxABCB19-2* (*ATP-binding cassette transporter B 19–2*) by binding to its promoter, forming a regulatory pathway that enhances cutting rooting efficiency and promotes primary root growth [29].

E. sibiricus, a heterologous tetraploid plant, is widely distributed across the Eurasian continent, exhibiting remarkable genetic diversity and strong ecological adaptability [30]. This is particularly valuable for establishing high-yield cultivated grasslands, especially on the Qinghai-Tibet Plateau and in northern China, as it addresses natural grassland degradation on one hand and alleviates livestock-forage imbalance on the other. Additionally, *E. sibiricus* plays a crucial role in restoring and improving natural grasslands, contributing to enhanced ecological conditions and increased grassland productivity [31]. Its high yield and resilience to environmental stressors make it an exceptional forage species, supporting the development of grassland-based livestock farming and ecological restoration efforts. Recently, the reference genome sequence of *E. sibiricus* was published, providing valuable insights into its genetic basis and evolutionary history through in-depth genome and population genomic analyses [32, 33]. This high-quality genomic research establishes a solid scientific foundation for future breeding programs and practical applications of *E. sibiricus*. The *SPL* gene family functions as a central regulatory hub governing plant growth and development, mediated by temporal-specific expression (via the miR156 module), multi-signal integration (hormonal and stress responses), and a balanced molecular mechanism of conservation and diversity. Its core advantage resides in maximizing biological outputs—including flowering, morphology, yield, and stress resistance—through minimal genetic investment (a single SBP domain). This positions *SPL* genes as pivotal targets for precision breeding in *E. sibiricus*.

In this study, the *SPL* genes were systematically explored, identified, and analyzed in *E. sibiricus* by using its genome as a reference. Evolutionary relationships were investigated through comparative analysis with *SPL* genes from seven representative species: *A. thaliana*, *Solanum lycopersicum*, *O. sativa*, *Sorghum bicolor*, *Zea mays*, *Brachypodium distachyon*, and *Triticum aestivum*. Additionally, chromosomal distribution, gene duplication events, cis-acting regulatory elements, gene structures, and conserved motifs of *EsSPL* family were investigated. Tissue-specific expression profiling of members was performed in roots, stems, leaves, spikes, seeds, and seedlings (including young roots) during the heading stage. Furthermore, expression responses to four abiotic stressors were analyzed to elucidate their roles in stress

adaptation mechanisms. In summary, this comprehensive analysis provided valuable insights into the *SPL* gene family in *E. sibiricus*, highlighting their crucial roles in growth, development and stress adaptation.

Materials and methods

Identification and chromosomal localization of *E. sibiricus* *SPL* genes

In this study, the genomic data for *E. sibiricus* was provided by the Scientific Data (<https://www.nature.com/articles/s41597-024-03622-4> provided on September 6th, 2024) [33]. The *A. thaliana* *AtSPL* sequences were retrieved from the TAIR database (<http://www.arabidopsis.org/>). To identify *EsSPL* genes, BLASTp analysis was performed in TBtools using *AtSPL* sequences as queries. Additionally, conserved domain analysis was conducted using the CDD-Search database (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>), while the InterPro database (<https://www.ebi.ac.uk/interpro/>) was utilized to screen for SBP-domain and non-SBP-domain proteins. Pfam model files were downloaded from Pfam database for querying sequences (<https://www.ebi.ac.uk/interpro/download/Pfam/>). Using these resources, TBtools confirmed that the *E. sibiricus* *SPL* gene family contains SBP domains (PF03110). The chromosomal localization of *EsSPL* genes was determined based on genome annotation data for *E. sibiricus*. Additionally, The physicochemical properties of *EsSPL* proteins were analyzed using the ExPASy protParam tool (<http://web.expasy.org/>). This analysis included parameters such as amino acid length, molecular weight (MW), isoelectric point (PI), instability index, aliphatic index, and average hydrophobicity (GRAVY).

Phylogenetic analysis of the *EsSPL* gene family

SPL amino acids sequences were obtained from multiple databases for *A. thaliana*, *S. lycopersicum*, *O. sativa*, *S. bicolor*, *B. distachyon*, *T. aestivum*, and *Z. mays*. These sequences were obtained from TAIR (The Arabidopsis Information Resource), the Rice Genome Annotation Project, the Sol Genomics Network, and GIGADB (Genomic Data Commons for Agriculture <http://www.gigadb.org>). To investigate the evolutionary relationships among the *EsSPL* gene family members, a comprehensive phylogenetic tree was constructed. For further functional analysis, additional *SPL* amino acid sequences from the same species were retrieved from NCBI (National Center for Biotechnology Information <https://www.ncbi.nlm.nih.gov/>), Pear MODB (a molecular database for plant sequences <http://www.plantgdb.org/>), Plant TFDB (a comprehensive database for plant transcription factors <http://planttfdb.gao-lab.org/>), and TAIR. The phylogenetic tree was generated using the Neighbor-Joining (NJ) method in MEGA 11.0 software, with 1,000 bootstrap

replications to ensure statistical robustness. The resulting tree was then visualized using the iTOL online platform (<http://itol.embl.de/>) to provide a clear and structured representation.

Gene structure and multiple sequence alignment analysis

Motif analysis of the *EsSPL* gene family proteins was conducted using the MEME tool (<http://meme-suite.org/tools/meme>) [18]. Additionally, the gene structures and conserved domains of *EsSPL* family members were analyzed using TBtools to gain a comprehensive understanding of their characteristics [34].

Gene duplication and synteny analysis of the *EsSPL* gene family

The chromosomal location and mapping information for the *EsSPL* gene family members were obtained from the *E. sibiricus* genome annotation file (GFF3 format) using TBtools. This enabled the visualization of gene distribution across chromosomes, as described in Chen's study [34]. To explore syntenic relationships, genomic sequences of *A. thaliana* and *T. aestivum* were downloaded from the Phytozome database (<https://phytozome-next.jgi.doe.gov/>). Synteny analysis was then performed using the One Step MCScanX function in TBtools, comparing *E. sibiricus* with *A. thaliana* and *T. aestivum*. The analysis parameters were set to a 'Num of Blast Hits' of 4 and an Expected value (*E*-value) threshold of 1×10^{-10} to ensure accuracy and reliability.

Identification of cis-acting regulatory elements in the promoters and prediction of protein-protein interactions

To investigate the potential functions and expression regulation mechanisms of the *EsSPL* genes, the 2000 bp upstream promoter sequences of these genes were extracted. The PlantCARE database (accessed on September 14, 2024, <http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) and TBtools were used for visualization [35]. Protein-protein interaction predictions were carried out using the online STRING database (accessed on September 14, 2024, <https://cn.string-db.org/>). Interaction protein information was retrieved from UniProt (<https://www.uniprot.org/>, accessed on September 14, 2024).

Growth conditions and stress treatments of *E. sibiricus*

The materials (number:16-118) of *E. sibiricus* utilized in this experiment were provided by the Academy of Animal Science and Veterinary in Qinghai University, China. These plant materials were cultivated within an artificial climate chamber at the Institute of Animal Science of CAAS in Beijing, China. Hydroponic seedlings of *E. sibiricus* were grown in a controlled environment with a photoperiod of 16 h light (25 °C) followed

by 8 h darkness (18 °C), a Light intensity of 250 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and a relative humidity of 70%, with regular maintenance. For abiotic stress treatments, seedlings two weeks old were utilized. To investigate gene expression levels in various tissues, roots, stems, leaves, and spikes were gathered at the heading stage for expression analysis. Additionally, seeds and young roots and seedlings of *E. sibiricus*, two weeks old, were sampled for expression analysis as well. All samples were subjected to three biological replicates. The data was visualized using Amazing Heat Map software to generate heatmaps that offer a clear representation of the experimental results. Regarding abiotic stress treatments, seeds of *E. sibiricus* were surface-sterilized and germinated on moist filter paper for 7 days. Uniform 7-day-old seedlings were then transferred to half-strength Murashige and Skoog (1/2 MS) hydroponic medium containing macro- (N, P, K, Ca, Mg) and micro-nutrients (I, B, S, Zn, Co, Cu, Mo), FeSO_4 , and organic supplements (myo-inositol, nicotinic acid, pyridoxine-HCl, thiamine, glycine), adjusted to pH 5.8. After 7 days of hydroponic culture, two weeks old seedlings were individually exposed to salt (200 mM NaCl), drought (20% PEG6000), abscisic acid (ABA at 0.1 mM), and gibberellin (GA at 0.1 mM) for the indicated durations [36]. Samples were collected at various time points: 0 h, 3 h, 6 h, 12 h, 24 h, 48 h, 72 h, and 100 h. All experimental procedures were conducted with three biological replicates to ensure accuracy and reliability.

Total RNA extraction and qPCR analysis

Total RNA was extracted utilizing the Trizol reagent (15596026 Invitrogen, Carlsbad, CA, USA), following the manufacturer's prescribed protocol. Subsequently, genomic DNA contamination was eliminated through DNase digestion employing the services of Promega (Madison, WI, USA). The synthesis of the first-strand cDNA was accomplished using M-MLV reverse transcriptase (R312-01 also provided by Vazyme) with 1 μg of the total RNA as a template. RT-qPCR analysis was performed on the CFX96 Touch™ Real-Time PCR Detection System from Bio-Rad (Hercules, CA, USA). The PCR amplification parameters were set as: an initial Denaturation at 95°C for 2 min, followed by 40 cycles of Denaturation at 95°C for 15 s and annealing/extension at 60 °C for 40 s. A final cycle consisted of Denaturation at 95°C for 15 s and annealing/extension at 60 °C for 15 s. The *E.sibiricus* actin gene was utilized as the internal reference, and relative gene expression was analyzed using the $2^{-\Delta\Delta\text{Ct}}$ methodology. All experimental procedures were repeated with three biological replicates and three technical replicates to ensure reliability. Primer Design was done using Primer 5.0 software for accurate and consistent results.

Data analysis

The statistical analysis was conducted utilizing the one-way analysis of variance (ANOVA) within the SPSS software version 23.0. The data was presented in the form of mean values accompanied by their corresponding standard Deviations. Graphs were created and thoroughly analyzed using GraphPad Prism 10.1.1 software. Any observed differences were deemed statistically significant when the *P*-value was below 0.05.

Results

Genome-wide identification of the SPL gene family in *E. sibiricus*

Using comparative genomics techniques, a total of 37 candidate genes were identified in the *E. sibiricus* genome [33]. These genes were named based on their sequence homology with corresponding *A. thaliana* SPL proteins. The key characteristics of the *EsSPL* gene family, including TIGR loci, chromosomal locations, coding and amino acid sequence lengths, molecular weights, theoretical isoelectric points, instability indices, aliphatic indices, and the hydrophilicity, are summarized in Table 1. The amino acid lengths of *EsSPL* proteins exhibit Significant variation, ranging from 192 to 1,129 residues. Their molecular weight span from 20,156.51 to 123,837.02. The theoretical isoelectric points value range from 5.49 to 9.99, while instability indices vary between 51.57 and 72.05, indicating that these proteins maybe unstable. Furthermore, the aliphatic index varies range from 41.47 to 80.84, reflecting differences in thermostability. Importantly, the average grand hydropathy (GRAVY) scores of *EsSPL* proteins are negative, suggesting that they are predominantly hydrophilic.

Phylogenetic tree analysis of the SPL gene family in *E. sibiricus*

To explore the evolutionary relationships of SPL proteins in *E. sibiricus*, a comprehensive phylogenetic analysis was conducted using multiple model plants species, including *A. thaliana*, *S. lycopersicum*, *O. sativa*, *S. bicolor*, *B. distachyon*, *T. aestivum*, and *Z. mays*. (Fig. 1). To further explore the structural complexity of *SPL* gene family, a Neighbor-Joining (NJ) phylogenetic tree was constructed based on SPL protein sequences from *E. sibiricus* (19), *A. thaliana* (17), *S. lycopersicum* (15), *O. sativa* (19), *S. bicolor* (18), *Z. mays* (31), *B. distachyon* (18), and *T. aestivum* (56). The phylogenetic analysis grouped these *SPL* genes into seven distinct clades, revealing evolutionary patterns among species. Notably, in Group 6, *EsSPL* genes clustered closely with wheat *TaSPL* genes, suggesting a strong co-evolutionary relationship between *E. sibiricus* and *T. aestivum*. Specifically, the homologous genes such as *EsSPL15*, *EsSPL16*, *EsSPL21*, *EsSPL22*, and *EsSPL23* exhibited high sequence similarity with

Table 1 List and detailed information of identified *SPL* genes in *Elymus sibiricus*

Gene Name	Gene ID	Chr	CDS Length	Pep Length	Molecular Weight(kDa)	Theoretical pl	Instability Index	Aliphatic Index	Grand Average of Hydrophobicity
<i>EsSPL1</i>	evm.model.Chr01.9142	1	579	192	20156.51	9.99	57.75	59.69	−0.627
<i>EsSPL2</i>	evm.model.Chr01.18876	1	966	321	33678.91	9.36	53.2	58.35	−0.459
<i>EsSPL3</i>	evm.model.Chr01.21724	1	1230	409	44107.57	7.79	57.36	49.02	−0.779
<i>EsSPL4</i>	evm.model.Chr02.9866	2	579	192	20175.47	9.87	62.39	58.65	−0.641
<i>EsSPL5</i>	evm.model.Chr02.18548	2	1386	461	47970.38	9.73	53.8	67.18	−0.254
<i>EsSPL6</i>	evm.model.Chr02.21396	2	1224	407	43930.34	7.05	57.01	48.8	−0.779
<i>EsSPL7</i>	evm.model.Chr03.12983	3	2580	859	94207.19	5.79	53.9	79.7	−0.31
<i>EsSPL8</i>	evm.model.Chr04.13338	4	2580	859	94063.92	5.7	51.57	80.84	−0.28
<i>EsSPL9</i>	evm.model.Chr05.4949	5	1557	518	55294.21	9.32	61.64	60.75	−0.454
<i>EsSPL10</i>	evm.model.Chr06.5732	6	1209	402	42334.95	9.04	58.85	54.75	−0.538
<i>EsSPL11</i>	evm.model.Chr07.7740	7	2514	837	92845.35	9.01	62.24	77.41	−0.388
<i>EsSPL12</i>	evm.model.Chr07.9472	7	1257	418	43461.13	8.85	55.22	50.98	−0.615
<i>EsSPL13</i>	evm.model.Chr07.9649	7	3390	1129	123837.02	7.06	54.04	72.93	−0.536
<i>EsSPL14</i>	evm.model.Chr07.9826	7	1095	364	38309.45	6.7	63.54	57.47	−0.493
<i>EsSPL15</i>	evm.model.Chr07.21378	7	1215	404	43488.89	9.16	60.22	50.89	−0.562
<i>EsSPL16</i>	evm.model.Chr07.21380	7	1257	418	44408.6	9.05	56	50.38	−0.609
<i>EsSPL17</i>	evm.model.Chr08.7950	8	2523	840	93125.69	8.54	59.21	80.62	−0.353
<i>EsSPL18</i>	evm.model.Chr08.9588	8	1146	381	39992.05	9.02	54.88	46.93	−0.755
<i>EsSPL19</i>	evm.model.Chr08.9908	8	3390	1129	123643.82	6.85	52.66	74.13	−0.511
<i>EsSPL20</i>	evm.model.Chr08.10315	8	1275	424	45158.16	6.75	60.14	57.88	−0.492
<i>EsSPL21</i>	evm.model.Chr08.20197	8	1197	398	43199.01	9.59	52.26	60.7	−0.487
<i>EsSPL22</i>	evm.model.Chr08.20224	8	1185	394	42515.77	8.9	60.12	52.44	−0.588
<i>EsSPL23</i>	evm.model.Chr08.20226	8	1251	416	44490.76	9.04	55.53	52.74	−0.604
<i>EsSPL24</i>	evm.model.Chr09.13491	9	1227	408	42487.41	8.39	61.73	41.47	−0.739
<i>EsSPL25</i>	evm.model.Chr09.14379	9	1290	429	44773.98	6.88	63.82	57.46	−0.378
<i>EsSPL26</i>	evm.model.Chr09.21556	9	2892	963	105347.61	5.57	54.13	78.71	−0.311
<i>EsSPL27</i>	evm.model.Chr10.13639	10	1203	400	41905.95	8.74	60.75	42.55	−0.705
<i>EsSPL28</i>	evm.model.Chr10.14363	10	1302	433	45246.59	6.88	61.07	55.8	−0.371
<i>EsSPL29</i>	evm.model.Chr10.21624	10	2898	965	105583.95	5.49	53.11	78.85	−0.301
<i>EsSPL30</i>	evm.model.Chr11.2280	11	1227	408	43789.34	8.71	60.82	60.59	−0.388
<i>EsSPL31</i>	evm.model.Chr11.5748	11	1422	473	51073.13	9.1	57.96	52.45	−0.619
<i>EsSPL32</i>	evm.model.Chr11.7263	11	1350	449	48342.73	7	72.05	47.95	−0.631
<i>EsSPL33</i>	evm.model.Chr11.7419	11	981	326	35316.67	9.34	67.48	53.9	−0.668
<i>EsSPL34</i>	evm.model.Chr12.4808	12	1422	473	51077.04	9.18	62.32	50.78	−0.645
<i>EsSPL35</i>	evm.model.Chr12.6316	12	1461	486	52010.72	8.59	63.11	63.15	−0.345
<i>EsSPL36</i>	evm.model.Chr12.6799	12	978	325	34954.24	9.34	61.04	54.4	−0.611
<i>EsSPL37</i>	evm.model.Chr14.18350	14	933	310	33707.14	6.82	61.52	56.94	−0.535

T. aestivum *TaSPL* genes (e.g., *TaSPL* 16A, *TaSPL*16D), implying potential functional conservation and shared evolutionary trajectories that warrant further investigation. Furthermore, the analysis revealed a significant expansion of the *SPL* gene family from lower to higher plants. This expansion is accompanied by extensive gene duplication events, which likely contributed to the evolutionary success of flowering plants by enhancing their adaptability to diverse and fluctuating environmental conditions [37].

Gene structure analysis of the SPL gene family in *E. sibiricus*

To further explore the functional diversity of *SPL*-associated candidate genes in *E. sibiricus*, we analyzed their conserved motifs, domains, and gene structures (Fig. 2). Using the MEME suite for multiple sequence alignments, the sequence markers that target the SBP domain in *E. sibiricus* were identified. This approach facilitated the comprehensive assessment of the conservation of homologous domain sequences, particularly in relation to the SBP domain. *SPL* proteins are structurally characterized by two zinc ion-binding motifs: C3H (Cys-Cys-Cys-His) and C2HC (Cys-Cys-His-Cys). Additionally, a Nuclear

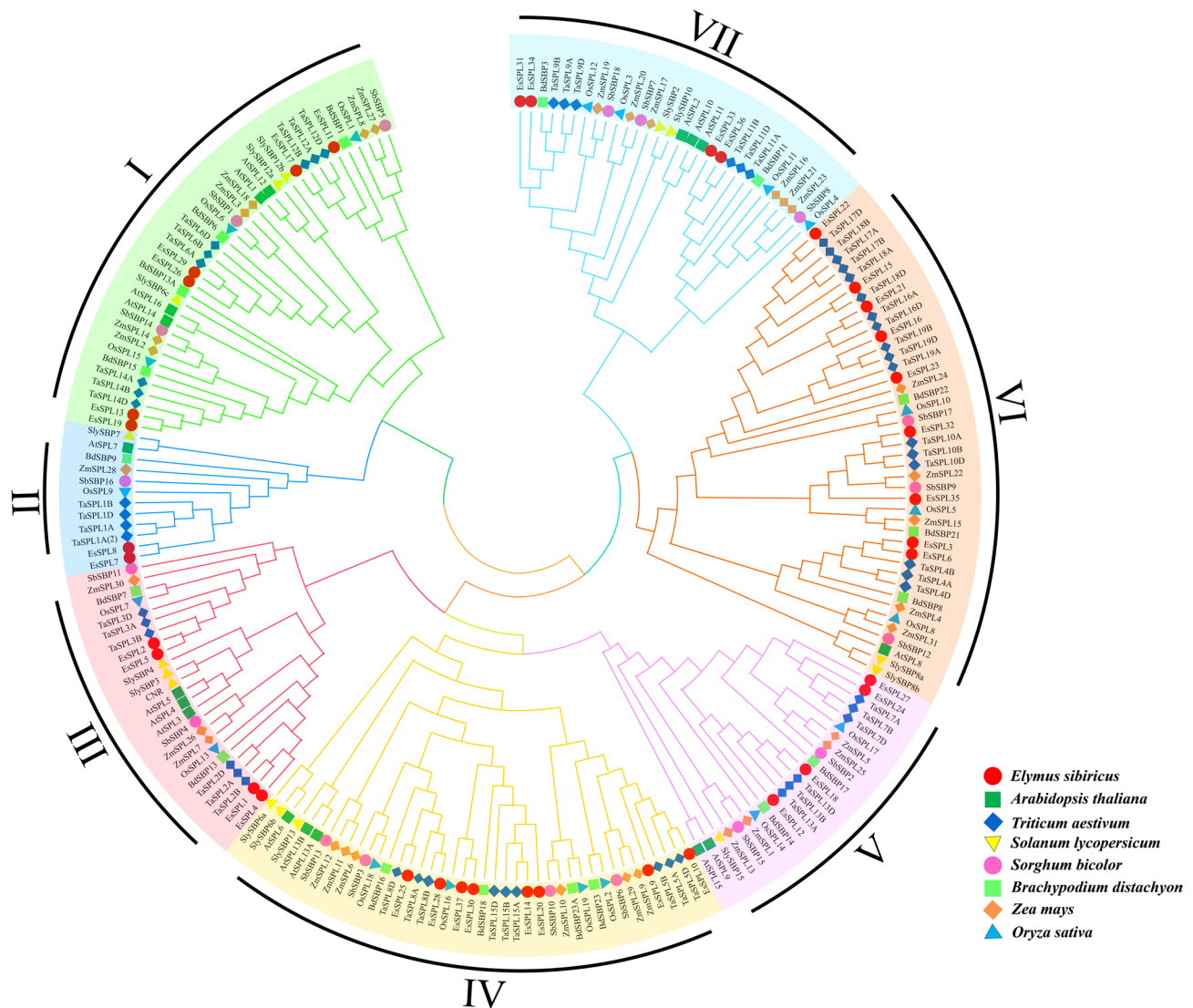


Fig. 1 Evolutionary analysis of the *SPL*-related candidate genes in *E. sibiricus*

Localization Signal (NLS) overlaps with the second zinc finger structure at the carboxyl terminus of the SBP domain. Our analysis revealed that motif 1 and motif 2 in *E. sibiricus* domain correspond to Zn finger motifs, both of which are present in all *EsSPL* proteins, highlighting a high degree of sequence conservation. Notably, *EsSPL37* is the only exception, as it lacks motif 2 (Fig. 2A). To investigate post-transcriptional regulation, we aligned complementary sequences of miR156 with the multiple sequence alignment of *EsSPL* genes (Supplementary Fig. 1). The analysis identified miR156 recognition Sites in 16 out of 37 *EsSPL* genes, accounting for 43.2% of the gene family. Interestingly, these recognition Sites were consistently located within motif 7, suggesting a conserved regulatory mechanism. Beyond the SBP domain, certain *EsSPL* proteins exhibited additional structural features. Specially, *EsSPL26* and *EsSPL29* possess an extra ANKYR

domain, which may imply functional diversification (Fig. 2B). Analysis of the intron-exon structures showed that the number of introns among *EsSPL* genes varies from 2 to 11, with a maximum of three exons (Fig. 2C). Collectively, these findings provide critical insights into the structural conservation and functional specificity of *SPL*-related candidate genes in *E. sibiricus*. The presence of conserved domains, motifs, and gene structures suggests their essential roles in various biological processes, highlighting their evolutionary significance and regulatory potential.

Chromosomal location and synteny analysis

Through mapping the open reading frames (ORFs) of all identified *EsSPL* genes to their respective chromosomes, the distribution of the *SPL* gene family in *E. sibiricus* were determined (Fig. 3A). The results indicated a

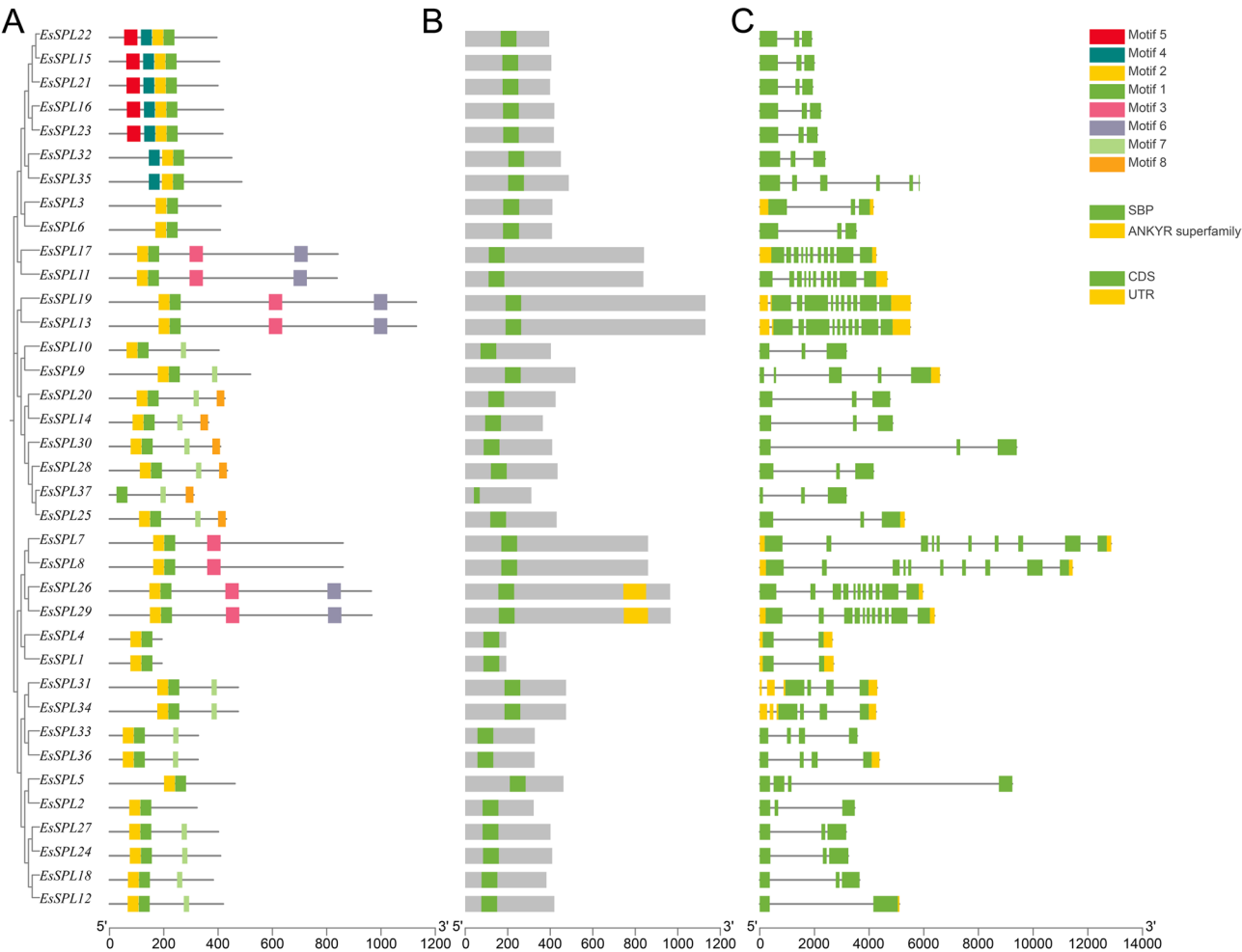


Fig. 2 Detailed information on the conserved motifs, functional domains, and exon-intron organization of *SPL*-related candidate genes in *E. sibiricus* where (A) represents conserved motifs, (B) represents functional domains, and (C) represents exon-intron organization

non-uniform distribution of *EsSPL* genes across the 14 chromosomes of *E. sibiricus*. Specifically, chromosome 8 contained the highest number of *SPL* genes, with 7 identified, followed by chromosomes 7 (6 genes), chromosome 11 (4 genes), and chromosome 1 and 2 (3 genes each). In contrast, chromosome 3, 4, 5, 6, and 14 contained only a single *EsSPL* gene, while chromosome 13 lacked any *EsSPL* genes entirely. Synteny analysis within *E. sibiricus* identified 19 gene pairs, demonstrating that members of the *EsSPL* gene family predominantly exist in tandem. This pattern is likely a consequence of the allopolyploid nature of *E. sibiricus*, which has led to the retention of duplicated genes (Fig. 3B). To further elucidate the evolutionary relationships of the *EsSPL* gene family, a comparative synteny map was constructed among *A. thaliana*, *E. sibiricus*, and *T. aestivum*. Interestingly, *A. thaliana* shares 6 orthologous genes pairs with *E. sibiricus*, whereas *E. sibiricus* shares a significantly higher number of orthologous genes pairs (25) with wheat (Fig. 3C).

Calculation of the nonsynonymous-to-synonymous substitution rate ratio (Ka/Ks) for paralogous gene pairs revealed that all values were less than 1, indicating that these genes are under purifying selection and have conserved functions throughout evolution (Table 2). Additionally, the estimated divergence times for both orthologous and paralogous gene pairs provide valuable evolutionary insights. The divergence time for orthologous gene pairs ranged from 37.83 to 168.38 million years ago (MYA), whereas paralogous gene pairs diverged between 36.84 and 246.61 MYA. These findings suggest that the expansion of the *SPL* gene family in *E. sibiricus* is likely a result of an ancient whole-genome duplication (WGD) event, followed by subsequent segmental duplications (Table 2).

Protein-protein interaction network analysis of the *SPL* gene family in *E. sibiricus*

A protein-protein interaction (PPI) network was constructed and analyzed using STRING network modeling

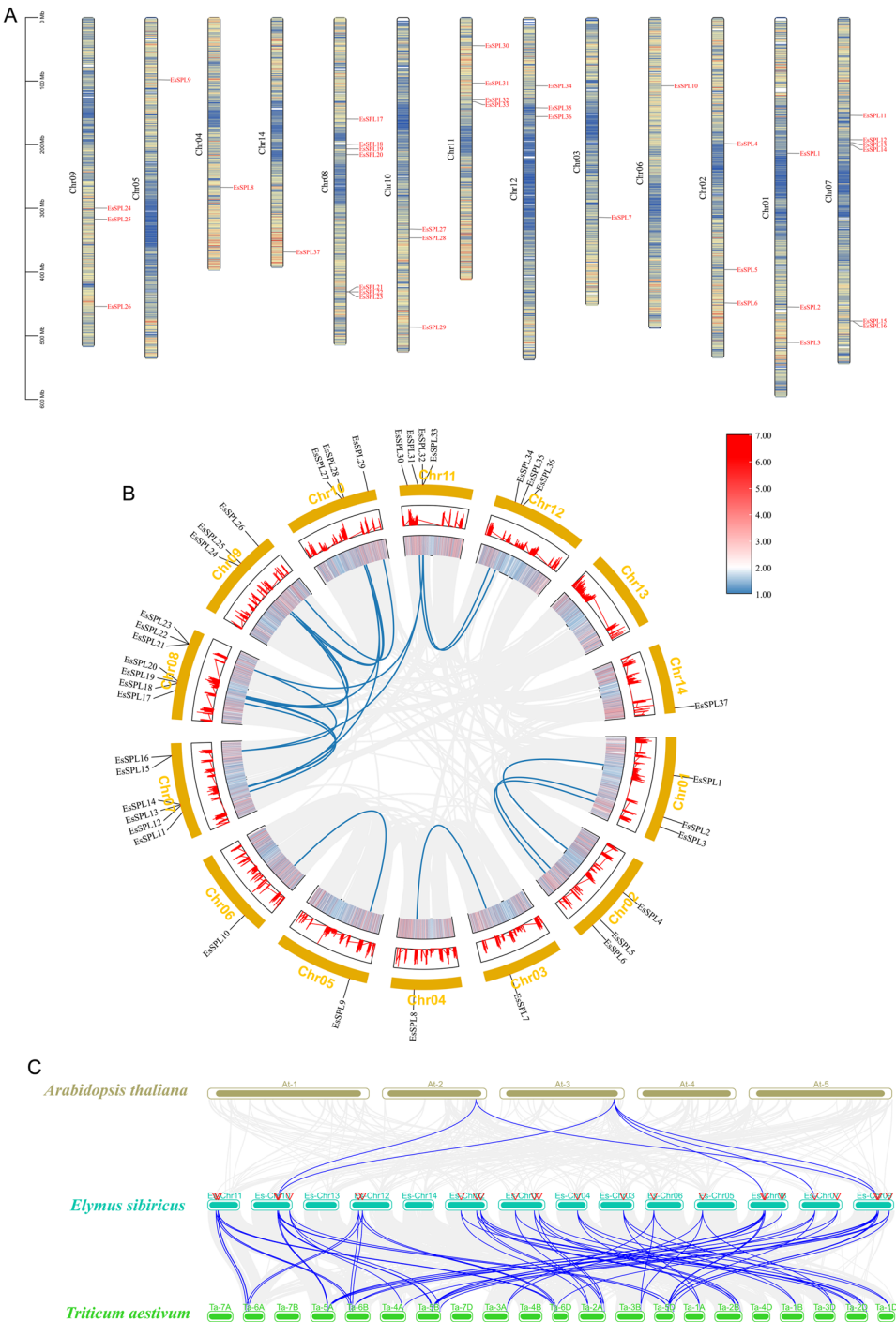


Fig. 3 Chromosomal distribution and synteny analysis of *SPL*-related candidate genes in *E. sibiricus*. **(A)** Spatial distribution of *SPL*-related candidate genes on the chromosomes of *E. sibiricus*. **(B)** Intra-species synteny analysis describing the chromosomal localization of *SPL*-related candidate genes. **(C)** Inter-species synteny analysis illustrating the relationship of *SPL*-related candidate genes among *A. thaliana*, *E. sibiricus* and wheat. Colored lines represent the syntenic relationships of *SPL*-related candidate genes between *E. sibiricus* and other plant species

technology, revealing key topological features. The analysis identified 37 *SPL* family proteins predicted to interact with a diverse range of proteins, suggesting their potential roles in pathways related to plant hormone signaling, environmental stress responses, as well as

growth and developmental processes (Fig. 4). Notably, homologous proteins exhibit similar interaction patterns, as observed in EsSPL1 and EsSPL4, EsSPL2 and EsSPL16, and various other EsSPL proteins pairs. This indicates that these protein pairs may share conserved

Table 2 Ka and Ks values of homologous gene pairs in *Elymus sibiricus*

	Ka	Ks	Ka/Ks	Purify selection	Duplication type	Time = Ks/2λ (MYAa)
EsSPL1/EsSPL2	0.2969	0.9044	0.3283	Yes	Segmental	74.74
EsSPL2/EsSPL9	0.3979	0.712	0.5588	Yes	Segmental	58.84
EsSPL3/EsSPL15	0.3754	0.8177	0.4591	Yes	Segmental	67.58
EsSPL4/EsSPL21	0.4454	0.8534	0.5219	Yes	Segmental	70.53
EsSPL12/EsSPL20	0.3821	1.0634	0.3593	Yes	Segmental	87.88
EsSPL15/EsSPL25	0.337	0.9228	0.3652	Yes	Segmental	76.26
EsSPL15/EsSPL33	0.4147	0.8941	0.4638	Yes	Segmental	73.89
EsSPL15/EsSPL36	0.4154	0.6889	0.6030	Yes	Segmental	56.93
EsSPL16/EsSPL20	0.3687	2.984	0.1236	Yes	Segmental	246.61
EsSPL20/EsSPL30	0.1366	0.5716	0.2390	Yes	Segmental	47.24
EsSPL22/EsSPL33	0.4147	0.9751	0.4253	Yes	Segmental	80.59
EsSPL23/EsSPL36	0.4277	0.8875	0.4819	Yes	Segmental	73.35
EsSPL24/EsSPL30	0.5296	2.4342	0.2176	Yes	Segmental	201.17
EsSPL30/EsSPL37	0.2636	0.4578	0.5758	Yes	Segmental	37.83

interaction domains, highlighting substantial connectivity within the *EsSPL* network and its intercatations with other proteins. Additionally, SOC1 and TOE2 frequently appear in these interaction networks. Previous studies have demonstrated that SOC1 gene functions as a central integrator of flowering signals, intergrating inputs from multiple pathways including photoperiod, autonomous, gibberellin, vernalization, and age-dependent pathways to regulate flowering time. Similarly, cytokinien modifies the transition from vegetative to reproductive growth via miR172 and its target genes TOE1 and TOE2 [38, 39]. This analysis provides deeper understanding of the functional regulation of interacting proteins within the *SPL* gene family, offering a comprehensive perspective on their intricate interactions and roles in biological processes.

Identification of cis-acting regulatory elements in the promoters of the SPL gene family in *E. sibiricus*

Using Plant CARE to analyze the promoter regions of *EsSPL* genes, a total of 966 cis-acting regulatory elements were identified (Fig. 5B). Among them, 369 cis-acting regulatory elements were associated with hormone response regulation (Fig. 5C). Four hormone-responsive motifs were detected, including methyl jasmonate response elements (CGTCA-motif), salicylic acid response elements (TCA-element), gibberellin response elements (P-box), and abscisic acid (ABA) response elements (ABRE). Additionally, various stress-responsive elements, were identified such as drought-related regulatory elements (MBS), low-temperature response elements (LTR), anaerobic response elements (ARE), and defense and stress response elements (TC-rich) etc. These cis-acting regulatory elements were classified into four major categories: environmental stress response elements, hormone response elements, development-related elements, and light response elements (Fig. 5B).

Among the environmental stress response elements, anaerobic response elements were the most abundant, accounting for 33.2% of the total. Within the hormone response element, ABRE motif were the most prevalent, constituting 64.8%, followed by CGTCA-motif elements at 27.4%. Among the five identified development-related elements, the CAT-box element was the most frequent((40.6%), followed by the O₂-site element (29.2%). For light-responsive elements, the G-box element was the most common(52.8%), followed by the Spl element (21.2%) (Fig. 5B). Notably, the distribution of cis-acting regulatory elements varied among different *EsSPL* genes. *EsSPL20* exhibited a higher proportion of hormone and light response cis-acting regulatory elements, whereas *EsSPL3* and *EsSPL6* contained a greater number of hormone response elements. In cntrast *EsSPL28* harbored more environmental stress response elements (Fig. 5C).

Tissue-specific expression of SPL-Related candidate genes in *E. sibiricus*

To investigate the potential functions of *SPL*-related candidate genes in different tissues of *E. sibiricus* and determine whether these genes serve broad housekeeping roles or paerticipate in tissue-specific regulatory mechanisms, a tissue-specific expression analysis was performed (Fig. 6). The results revealed that *EsSPL33* exhibits strong expression in root tissues, while *EsSPL15* and *EsSPL16* are highly expressed in stem. Similarly, *EsSPL4* demonstrates elevated expression in leaves. Several genes, including *EsSPL8*, *EsSPL10*, *EsSPL25*, *EsSPL28*, *EsSPL34*, and *EsSPL36*, displayelevated expression levels in spike tissues. Notably, a large subset of *EsSPL* genes (*EsSPL1*, *EsSPL2*, *EsSPL3*, *EsSPL6*, *EsSPL7*, *EsSPL13*, *EsSPL14*, *EsSPL17*, *EsSPL18*, *EsSPL21*, *EsSPL24*, *EsSPL26*, *EsSPL29*, *EsSPL31*, *EsSPL35*, and *EsSPL37*) showed predominant expression in seedlings. Additionally, *EsSPL9*, *EsSPL23*, *EsSPL27*, and *EsSPL35* exhibit strong expression in young

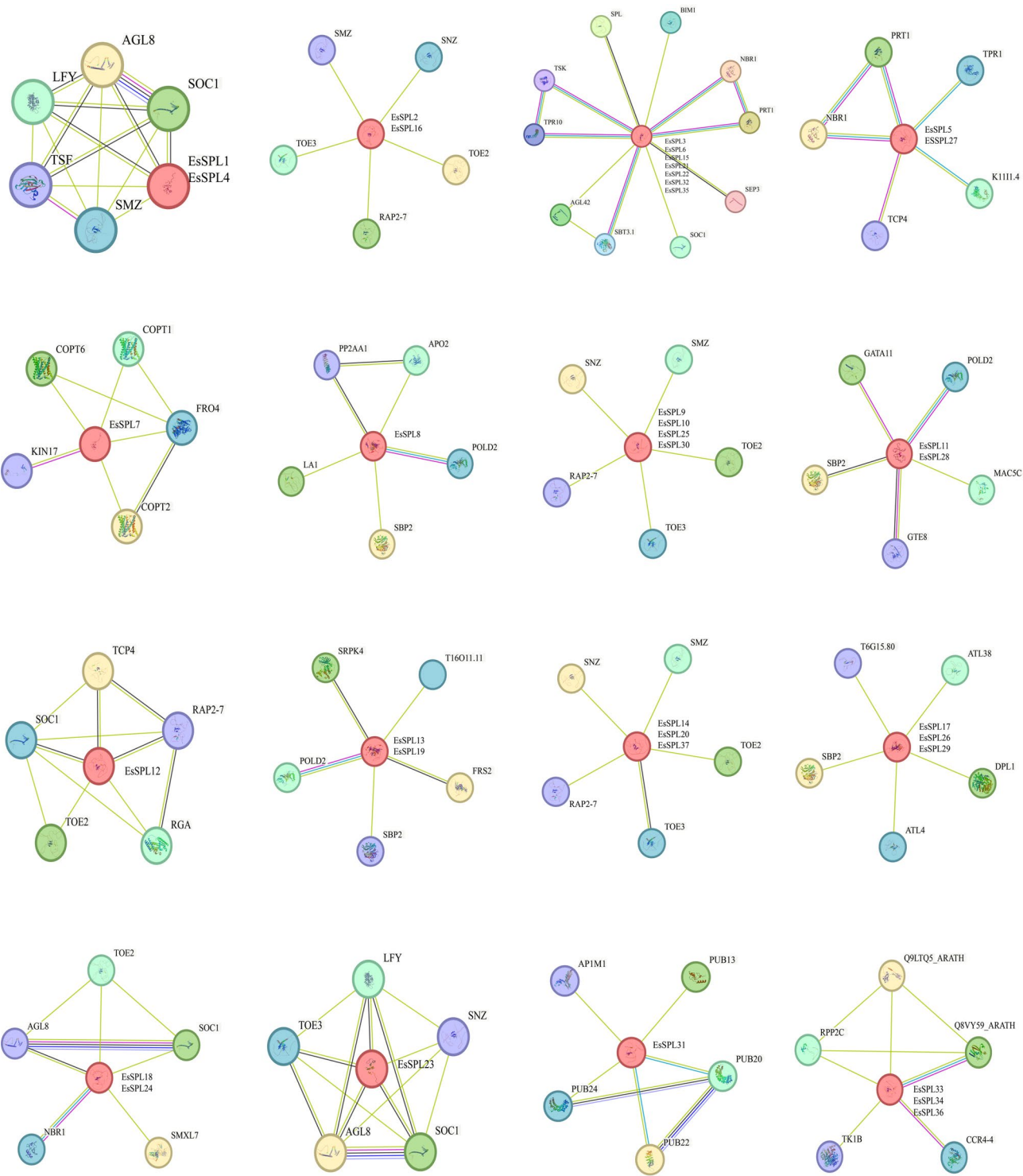


Fig. 4 Predicted regulatory network of *SPL* and its interacting proteins. Sky blue lines represent data from selected databases; purple lines are experimentally validated interactions; dark blue lines indicate gene co-occurrence; green lines represent text mining results; black lines denote co-expression; light blue lines indicate protein homology

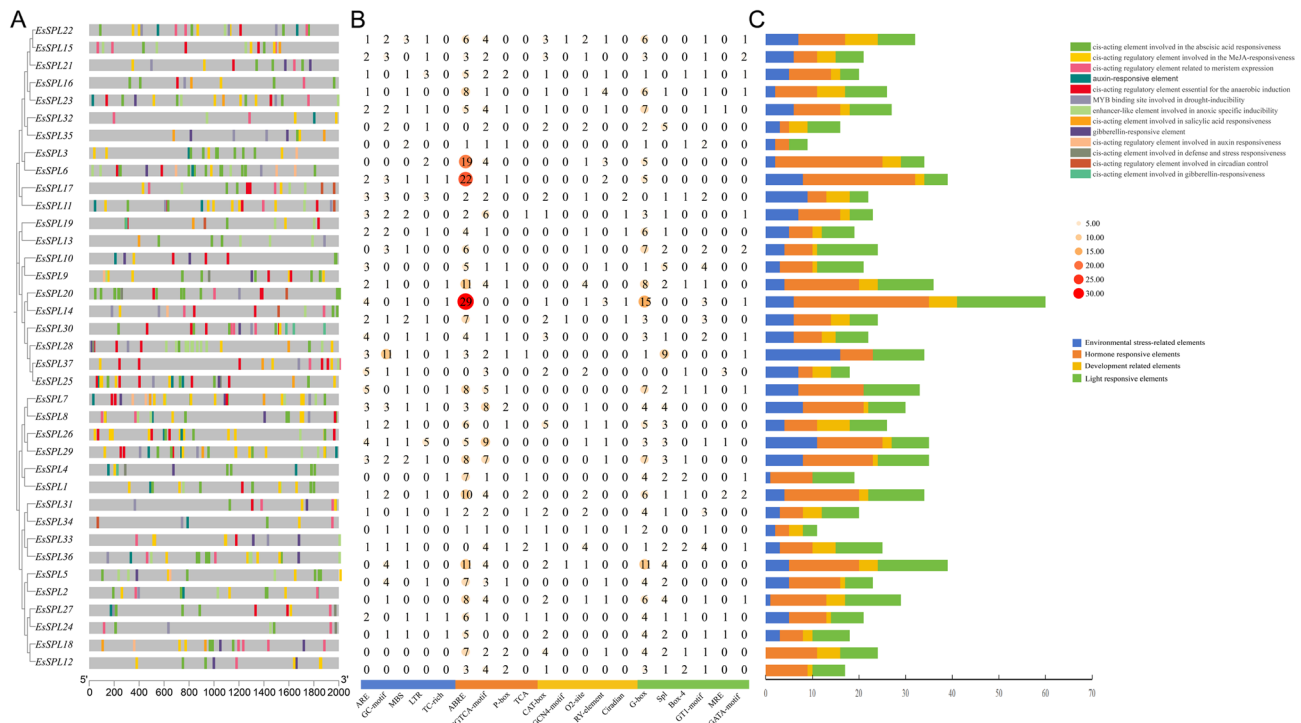


Fig. 5 Quantitative analysis of the number of cis-acting regulatory elements in *SPL*-related candidate genes. Different color shades and corresponding grid numbers represent the number of distinct promoter elements identified in the *SPL*-related candidate genes. The multicolored histogram shows the total number of cis-acting regulatory elements in each functional category: blue represents environment stress-related elements, orange represents hormone response elements, yellow represents growth and development-related elements, and green represents light response elements

roots, whereas *EsSPL5*, *EsSPL11*, *EsSPL12*, *EsSPL19*, *EsSPL20*, *EsSPL22*, *EsSPL30*, and *EsSPL32* are highly expressed in seeds. It is worth noting that, similar to *Setaria italic* (foxtail millet), most *SPL* gene family members in *E. sibiricus* exhibit high expression levels in seedlings, suggesting a conserved regulatory role during plant early development stages.

Expression of *E. sibiricus* *SPL* candidate genes under abiotic stress (salt and drought) and hormone treatments (ABA and GA)

To assess the role of *SPL*-related candidate genes in the stress response of *E. sibiricus*, their expression levels were analyzed following exposure to abiotic stress and hormone treatments conditions. two weeks old plants were subjected to hydroponic treatments, 200 mM NaCl (salt stress), 20% PEG6000(drought stress), 0.1 mM abscisic acid (ABA), and 0.1 mM gibberellin (GA). Treatments were administered for up to 100 h, with samples collected at intervals of 3, 6, 12, 24, 48, 72, and 100 h. For each treatment and time point, three plants were carefully harvested, and gene expression was quantified using three biological replicates to ensure accuracy under salt stress conditions, most *SPL* genes exhibited a down-regulation trend over time. Specifically, *EsSPL5*, *EsSPL6*, *EsSPL13*, *EsSPL16*, *EsSPL19*, *EsSPL25*, *EsSPL26*, *EsSPL27*,

and *EsSPL28* were significantly downregulated compared to their pre-treatment expression levels. In contrast, *EsSPL2*, *EsSPL7*, *EsSPL10*, *EsSPL11*, *EsSPL14*, *EsSPL17*, *EsSPL20*, *EsSPL21*, *EsSPL23*, *EsSPL33*, and *EsSPL34* exhibited an initial increase followed by a decline. Notably, *EsSPL4* and *EsSPL9* genes showed sustained upregulation, with expression levels increasing 5.45-fold and 6.55-fold, respectively, compared to CK-treatment levels (Fig. 7). These findings suggested that *EsSPL4* and *EsSPL9* may play significant roles in negative regulation under salt stress and could serve as potential targets for future studies on salt stress adaptation. Similarly, under drought stress conditions, many *SPL* genes exhibited an initial upregulation followed by a gradual decrease over time. Specifically, *EsSPL2*, *EsSPL3*, *EsSPL10*, *EsSPL11*, *EsSPL12*, *EsSPL13*, *EsSPL14* and *EsSPL18*, were significantly upregulated. Among them, *EsSPL8* and *EsSPL16* genes demonstrated the most pronounced increases, with expression levels rising 5.66-fold and 4.75-fold, respectively, compared to CK treatment levels (Fig. 7). These results indicated that *SPL* gene family members play a crucial role in drought stress regulation, particularly *EsSPL8* and *EsSPL16*, which may contribute to osmotic stress adaptation and root development processes.

Under ABA treatment, the expression levels of *EsSPL17*, *EsSPL22*, and *EsSPL36* gradually decreased

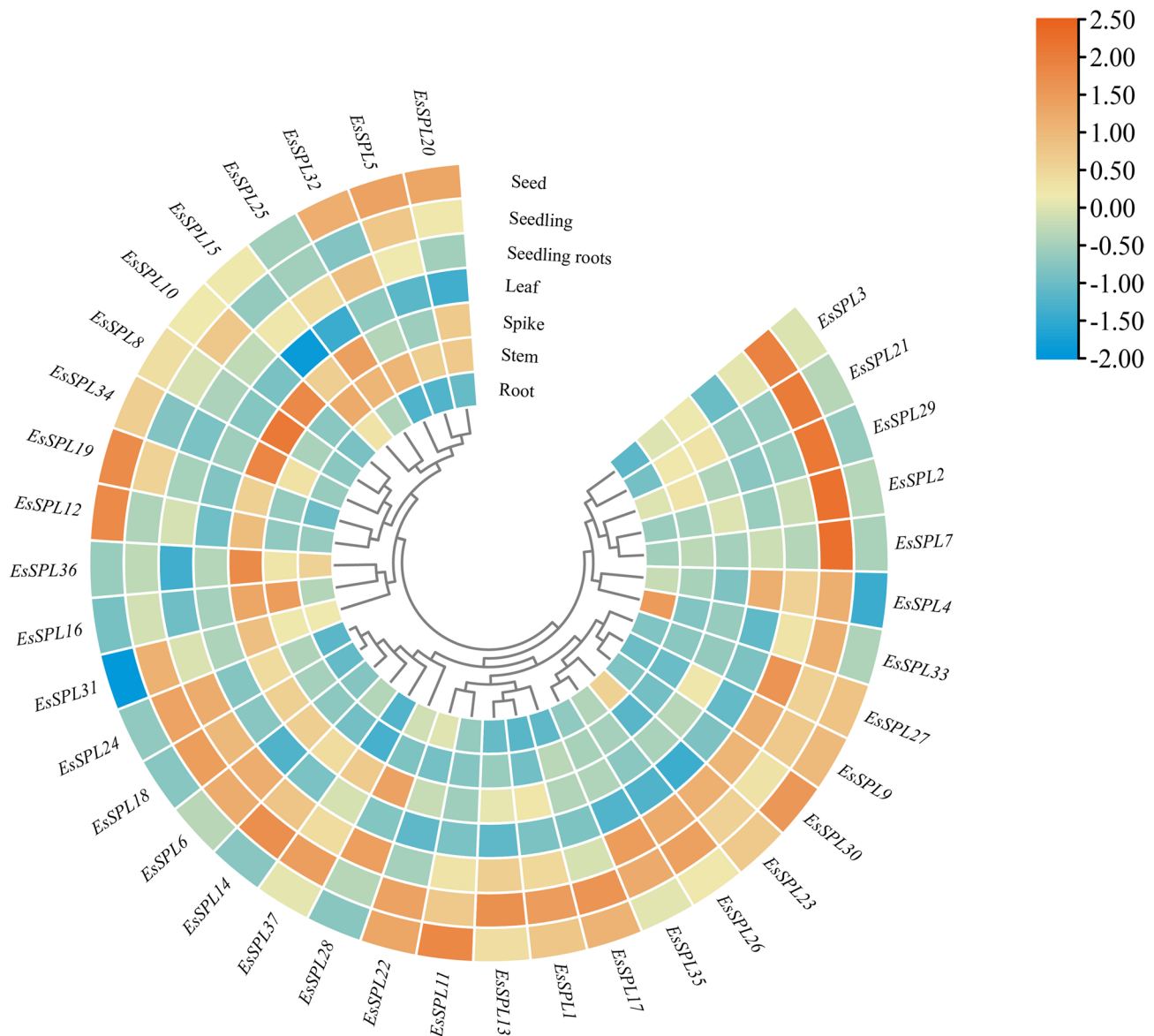


Fig. 6 Expression levels of *SPL*-related candidate genes in different tissues

over time, indicating significant downregulation compared to CK treatment levels. In contrast, *EsSPL6*, *EsSPL11*, *EsSPL12*, *EsSPL15*, *EsSPL27*, *EsSPL30*, *EsSPL31*, *EsSPL33*, and *EsSPL34* exhibited an initial upregulation followed by a subsequent decline (Fig. 8). Specifically, *EsSPL2* displayed a substantial regulation of 5.52-fold compared to CK treatment levels. Interestingly, the upregulation of *EsSPL2*, *EsSPL11*, *EsSPL12*, and *EsSPL27* under ABA treatment was consistent with their expression patterns observed in the drought treatment, suggesting that *SPL* transcription factors may play a crucial role in ABA-mediated stress responses. GA (gibberellin) regulates various plant growth and developmental processes, including seed germination, flower organ development, etc. Under GA treatment, the expression

levels of several genes including *EsSPL7*, *EsSPL8*, *EsSPL9*, *EsSPL10*, *EsSPL15*, *EsSPL16*, *EsSPL19*, *EsSPL20*, *EsSPL22*, *EsSPL25*, *EsSPL27*, *EsSPL32*, *EsSPL34*, *EsSPL36*, and *EsSPL37* initially increased before gradually declining, indicating a significant upregulation in comparison to their CK treatment levels. Specifically, *EsSPL2*, *EsSPL14*, and *EsSPL17* exhibited the most pronounced increases, with expression levels rising 3.92-fold, and 5.01-fold, and 6.31-fold, respectively (Fig. 8). These findings suggested that *SPL* might be involved in gibberellin signaling transduction [39], further reinforcing their regulatory roles in plant growth and stress adaptation.

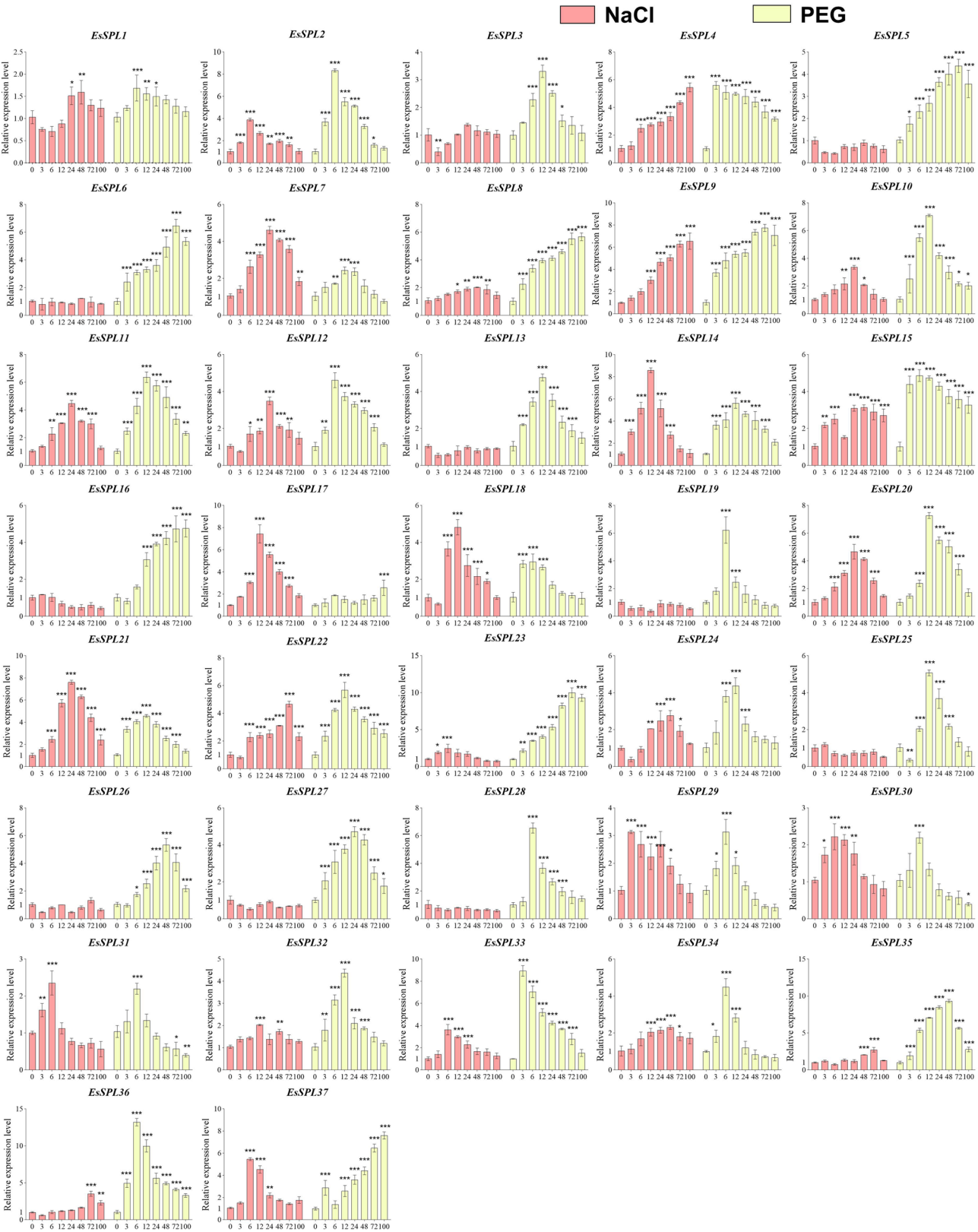


Fig. 7 Expression of *SPL*-related candidate genes under salt and drought stress treatments. The relative expression levels of these genes were quantified at specific time points: 0, 3, 6, 12, 24, 48, 72, and 100 h for each treatment. The data presented are means $\pm$ standard deviation (SD) calculated from three independent biological replicates. To determine statistical significance, a one-way analysis of variance was conducted, and the results are indicated as follows: “*” denotes $P < 0.05$; “**” denotes $P < 0.01$; and “***” denotes $P < 0.001$, $n = 6$



Fig. 8 Expression of *SPL*-related candidate genes under ABA and GA treatments. The relative expression levels of these genes were quantified at specific time points: 0, 3, 6, 12, 24, 48, 72, and 100 h for each treatment. The data presented are means $\pm$ standard deviation (SD) calculated from three independent biological replicates. To determine statistical significance, a one-way analysis of variance was conducted, and the results are indicated as follows: “*” denotes $P < 0.05$; “**” denotes $P < 0.01$; and “***” denotes $P < 0.001$, $n = 6$

Discussion

In this study, a total of 37 *SPL*-related candidate genes were identified in the genome of *E. sibiricus* and categorized into eight distinct groups. Comparative analysis with other plant species reveals a diverse gene distribution, including 17 *SPL* genes from *A. thaliana* [40], 15 from *S. lycopersicum* [41], 17 from *S. bicolor* [42], 19 from *O. sativa* [43], and a notably larger count of 31 from *Z. mays* [44]. Evolutionary analysis, gene structure examination, and sequence alignment demonstrated classification patterns in *E. sibiricus*, similar to those observed in other species. All *EsSPL* genes contained the SBP domain, while *EsSPL26* and *EsSPL29* also possessed an Ankyrin domain, consistent with previous findings on the *SPL* gene family structure in lychee [16].

Gene duplication is a key driver of genetic evolution [45]. In *E. sibiricus*, an evolutionary pattern similar to that of *T. aestivum* was observed. While most *EsSPL* genes exist as single-copy genes, several multi-copy genes were identified, including *EsSPL1*, *EsSPL4*, *EsSPL7*, and *EsSPL8*, as well as *EsSPL12*, *EsSPL16*, *EsSPL20*, *EsSPL24*, and *EsSPL30*, which are also found across various species. Notably, duplicated *SPL* genes in *E. sibiricus* are distributed across different chromosomes (Fig. 3), reflecting their complex evolutionary history. Most *SPL* regulatory genes appear to have originated from whole-genome duplication (WGD) events, while genes such as *EsSPL4*, *EsSPL8*, *EsSPL16*, *EsSPL20*, and *EsSPL30* likely arose through segmental duplications (Fig. 3). These findings suggested that the expansion of *SPL* genes is predominantly driven by WGD and segmental duplication. Moreover, interspecies collinearity analysis revealed that 67.57% of *EsSPL* genes have orthologs in wheat; whereas only 16.22% share orthology with *A. thaliana*, highlighting the increasing functional divergence of *SPL* genes between monocots and dicots. To further explore the evolutionary dynamics of *EsSPL* genes, synonymous (*Ks*) and nonsynonymous (*Ka*) substitution rates were calculated for paralogous gene pairs. The *Ka/Ks* ratio, an indicator of selective pressure [46] was examined: *Ka/Ks* ratio > 1 indicates positive selection, *Ka/Ks* = 1 suggests neutral selection, *Ka/Ks* < 1 indicates purifying selection. The results showed an average *Ka/Ks* ratio of 0.41 for paralogous *EsSPL* gene pairs in *E. sibiricus*, indicating that the *EsSPL* genes have undergone purifying selection and have been subject to strong evolutionary constraints.

In plants, the exon/intron structure of the *SPL* gene exhibits low conservation with exon numbers varying significantly. In *E. sibiricus*, *EsSPL* genes contain 2 to 11 exons (Fig. 2), a range comparable to that in chickpea (2–10 exons) [47, 48], sunflower (2–11 exons) [49], and lychee (2–10 exons) [16]. Interestingly, homologous *EsSPL* often differ in exon count, such as *EsSPL32* (3 exons) and *EsSPL33* (6 exons), suggesting that exon

gain or loss has contributed to structural variations during evolution. Despite similar motif sequences and arrangements among *EsSPL* genes (Fig. 2), substantial motif diversity was observed. Motif 3 and motif 6 are exclusive to the fifth evolutionary group, while motif 4 and motif 5 are unique to the sixth group. Additionally, motif 8 is found only in *EsSPL14*, *EsSPL20*, *EsSPL25*, *EsSPL28*, *EsSPL30*, and *EsSPL37*. Conversely, the highly conserved motifs 1 and 2, present in the core regions of *EsSPL* proteins, likely play a crucial role in their function as transcription factors [50]. Consequently, the structural diversity of *EsSPL* proteins is closely linked to their functional divergence, with additional motifs contributing to their specialized roles in *E. sibiricus*.

This study analyzes the promoters of *SPL*-related candidate genes, identifying 19 cis-acting regulatory elements associated with environmental stress, plant growth, development, hormone response, and light response (Fig. 5). Universally prevalent elements include the hypoxia-specific response element (ABRE), linked to meristematic cell division and proliferation (CAT box), light-responsive elements (MRE, GT1-motif). Specifically, ABRE and CAT box elements correlate with meristematic tissue development and hypoxic stress responses [5]. Research by Weits et al. [51] and Shukla et al. [52] highlighted the significance of apical meristem development under low oxygen conditions, essential for initiating new leaf formation. Moreover, hypoxia affects lateral root primordia by inhibiting the degradation of key regulatory proteins. Under such conditions, light acts as a signaling factor, activating stem cells through CK signaling and metabolic pathways [53]. The abundance of developmental elements (ABRE, CAT-box) in *SPL* promoters suggests their involvement in plant hormone signaling [54, 55], particularly in response to GA, which influences plant height and morphology. Furthermore, key cis-acting regulatory elements (GT-1-motif, CAT-box, GATAT-motif) have been identified in *E. sibiricus* *SPL* gene promoters [56]. *SPL* transcription factors regulate flowering in a tandem manner in *E. sibiricus* and serve as a target of miR156. Overexpression of *OsmiR156b* and *OsmiR156h* in previous studies resulted in dwarfism, reduced inflorescence size, and delayed flowering, emphasizing the pivotal role of miR156-*SPL* regulation in embryonic development [57]. Given the importance of *SPL* expression in flowering, future studies should focus on the functional validation of these cis-acting regulatory elements, exploring *SPL* interactions to uncover novel insights into plant growth, morphology, and developmental processes.

In organisms, gene expression precedes and is essential for gene function, with expression patterns intricately linked to gene functions [57]. The *SPL* gene family is widely involved in plant growth and development, playing a key role in regulating traits from seedlings to

maturity [58, 59]. In *A. thaliana*, *AtSPL10* eliminates de novo shoot regeneration capacity by derepressing the attenuation of cytokinin response in the primary root [19]. In *Rhododendron molle*, *RmSPL1/7/8/12/13* exhibit significantly divergent expression, suggesting their essential roles in regulating floral bud differentiation [60]. *TaSPL6B* recruits *TaD53* and *TaSPL3* to integrate light and strigolactone signaling pathways, thereby suppressing the key tillering inhibitor *TaTB1* and enhancing *TaD53*-mediated inhibition of *TaSPL3* in *T. aestivum* [61]. Similarly, *SPL* genes also play crucial roles in abiotic stress responses. This study demonstrates that under salt stress, *EsSPL4* and *EsSPL9* exhibit sustained upregulation and likely function as important negative regulators, suggesting their potential as key targets for future research on salt stress adaptation and genetic enhancement strategies. In studies on salt tolerance in *Codonopsis pilosula*, *CpSPL5* and *CpSPL8* were identified as negative regulators of salt stress resistance, indicating that members of the *SPL* gene family are involved in the plant SOS pathway [62]. In *Arachis hypogaea* (peanut) salt tolerance studies, *AhSPL5*, *AhSPL16*, *AhSPL25*, and *AhSPL36* were found to be upregulated under both drought and salinity conditions [63]. In this study, *EsSPL8* and *EsSPL16* showed the most significant unregulation under drought stress. However, no consistent expression patterns were observed between salt and drought treatments, which might be attributed to differences in stress application conditions. In *Mangifera indica* (mango) stress resistance studies, overexpression of *MiSPL3a* and *MiSPL3b* in transgenic *A. thaliana* conferred enhanced drought tolerance and ABA-induced stress resistance [64]. Under ABA treatment, *EsSPL2*, *EsSPL11*, *EsSPL12*, and *EsSPL27* displayed expression patterns consistent with those observed under drought stress, suggesting that *SPL* transcription factors play pivotal roles in both drought and ABA-mediated stress responses. Under GA treatment, *EsSPL2*, *EsSPL14*, and *EsSPL17* exhibited the most pronounced upregulation, indicating a potential role in gibberellin signal transduction and further emphasizing their regulatory functions in plant growth and stress adaptation [65]. Both GA and *SPL* genes regulate the key flowering integrators such as *LFY* and *SOC1*. GA promotes flowering by alleviating *DELLA*-mediated repression of these genes, while *SPL* transcription factors directly activate their transcription. This synergistic regulatory mechanism enables plants to fine-tune flowering time in response to internal developmental cues and external environmental signals, thereby ensuring reproductive success.

Based on these findings, we propose that the *SPL* gene family plays a crucial role in *E. sibiricus* abiotic stress resistance. Specifically, *EsSPL2*, *EsSPL4*, *EsSPL8*, *EsSPL9*, *EsSPL14*, *EsSPL16*, and *EsSPL17* may function as

key mediators of stress tolerance. Notably, *EsSPL2* gene consistently exhibited high expression across all stress treatments compared to the control (CK), under the four stress conditions examined.

Conclusion

This study represents the first genome-wide investigation of the *SPL* gene family in *E. sibiricus*, identifying 37 *EsSPL* members primarily expanded through segmental duplications. Regulatory analysis revealed the presence of hormone and stress-responsive cis-acting elements in *EsSPL* promoters, while protein interaction predictions implicated *SOC1* and *TOE2* as key interacting partners. Expression profiling highlighted tissue-specific expression patterns in seedlings, with *EsSPL2* exhibiting both constitutive expression and significant induction under four abiotic stresses. In summary, this research enhances the understanding of the genetic evolution and biological functions of the *EsSPL* gene family, providing a foundation for future functional characterization and application in stress adaptation studied.

Abbreviations

SPL	Squamosa promoter binding protein-like
IPA1	Ideal Plant Architecture 1
MW	Molecular weight
PI	isoelectric point
ABA	Absciscic acid
GA	Gibberellin

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-07366-0>.

Supplementary Material 1: Supplementary table 1. Classification of promoter cis-elements. Supplementary table 2. Organization of promoter cis-elements. Supplementary table 3. Primer design for *SPL* gene family.

Supplementary Material 2: Supplementary fig. 1. Multiple sequence alignment of the miR156 complementary sequences with the *SPL* genes of *E. sibiricus*.

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Authors' contributions

Performed the experiments with the help of J.T., K.X., Z.J., L.M., X.M., T.M., C.L., D.Y., F.L., and W.L. Analyzed the sequencing data; X.W., J.T., D.Y., W.L., M.H., J.H., and X.W. Designed the experiments, interpreted the results, and wrote the manuscript. All authors read and approved of the final manuscript.

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Data availability

The datasets supporting the results of this article are included in the article and Additional files.

Supplementary information**Ethics approval and consent to participate**

This article does not contain any studies involving human participants or animals performed by the authors. These methods were carried out by relevant guidelines and regulations. *E. sibiricus* used in this study is a prevalent wild species in the Qinghai area. Based on international standards, all the experimental research and field studies on plants, including the collection of plant material were approved by Institute of Animal Sciences. These materials are stored in Institute of Animal Sciences of Chinese Academy of Agricultural Sciences.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Fan E, Liu C, Wang Z, Wang S, Ma W, Lu N, et al. Genome-wide identification and expression analysis of the SQUAMOSA promoter-binding protein-like (SPL) transcription factor family in *Catalpabungei*. *Int J Mol Sci*. 2023;20(1):97.
- Jiang M, He Y, Chen X, Zhang X, Guo Y, Yang S, et al. CRISPR-based assessment of genomic structure in the conserved SQUAMOSA promoter-binding-like gene clusters in rice. *Plant J*. 2020;104(5):1301–14.
- Preston JC, Hileman LC. Functional evolution in the plant SQUAMOSA-promoter-binding protein-like (SPL) gene family. *Front Plant Sci*. 2013;4:80.
- He B, Gao S, Lu H, Yan J, Li C, Ma M, et al. Genome-wide analysis and molecular dissection of the SPL gene family in *Fraxinus mandshurica*. *BMC Plant Biol*. 2022;21(1):451.
- Ibraheem O, Botha CE, Bradley G. In silico analysis of cis-acting regulatory elements in 5' regulatory regions of sucrose transporter gene families in rice (*Oryza sativa* Japonica) and *Arabidopsis Thaliana*. *Comput Biol Chem*. 2010;34(5–6):268–83.
- Schwarz S, Grande AV, Bujdosó N, Saedler H, Huijser P. The MicroRNA regulated SBP-box genes *SPL9* and *SPL15* control shoot maturation in *Arabidopsis*. *Plant Mol Biol*. 2008;67(1–2):183–95.
- Gandikota M, Birkenbihl RP, Höhmann S, Cardon GH, Saedler H, Huijser P. The miRNA156/157 recognition element in the 3' UTR of the *Arabidopsis* SBP box gene *SPL3* prevents early flowering by translational inhibition in seedlings. *Plant J*. 2007;49(4):683–93.
- Wang JW, Czech B, Weigel D. miR156-regulated *SPL* transcription factors define an endogenous flowering pathway in *Arabidopsis Thaliana*. *Cell*. 2009;138(4):738–49.
- Wang ZM, Xue W, Dong CJ, Jin LG, Bian SM, Wang C, et al. A comparative miRNAome analysis reveals seven fiber initiation-related and 36 novel miRNAs in developing cotton ovules. *Mol Plant*. 2012;5(4):889–900.
- Tripathi RK, Bregitzer P, Singh J. Genome-wide analysis of the *SPL*/miR156 module and its interaction with the AP2/miR172 unit in barley. *Sci Rep*. 2018;8(1):7085.
- Huijser P, Schmid M. The control of developmental phase transitions in plants. *Development*. 2011;138(19):4117–29.
- Cui L, Zheng F, Wang J, Zhang C, Xiao F, Ye J, et al. miR156a-targeted SBP-Box transcription factor *SISPL13* regulates inflorescence morphogenesis by directly activating SFT in tomato. *Plant Biotechnol J*. 2020;18(8):1670–82.
- Sun Y, Wang Y, Xiao Y, Zhang X, Du B, Turupu M, et al. Genome-wide identification of the SQUAMOSA promoter-binding protein-like (SPL) transcription factor family in sweet cherry fruit. *Int J Mol Sci*. 2023;24(3):2880.
- Jiang Y, Liu C, He G, Zhang Y, Liu M, Zhang K, et al. Regulation of ginseng adventitious root growth in *Panax ginseng* by the miR156-targeted *PgSPL24-09* transcription factors. *Plant Physiol Biochem*. 2024;215:109026.
- Lian H, Wang L, Ma N, Zhou CM, Han L, Zhang TQ, Wang JW. Redundant and specific roles of individual MIR172 genes in plant development. *PLoS Biol*. 2021;19(2):e3001044.
- Xu Z, Wu J, Jing X, Khan FS, Chen Y, Chen Z, et al. Genome-wide identification of Litchi *SPL* gene family and expression analysis in pericarp anthocyanin biosynthesis. *Horticulturae*. 2024;10(7):762.
- Wang Z, Wang Y, Kohalmi SE, Amyot L, Hannoufa A. SQUAMOSA promoter-binding protein-like 2 controls floral organ development and plant fertility by activating asymmetric leaves 2 in *Arabidopsis Thaliana*. *Plant Mol Biol*. 2016;92(6):661–74.
- Shikata M, Koyama T, Mitsuda N, Ohme-Takagi M. *Arabidopsis* SBP-box genes *SPL10*, *SPL11* and *SPL2* control morphological change in association with shoot maturation in the reproductive phase. *Plant Cell Physiol*. 2009;50(12):2133–45.
- Barrera-Rojas CH, Rocha GHB, Polverari L, Pinheiro-Brito DA, Batista DS, Notini MM, et al. miR156-targeted *SPL10* controls *Arabidopsis* root meristem activity and root-derived de novo shoot regeneration via cytokinin responses. *J Exp Bot*. 2020;71(3):934–50.
- Wu G, Poethig RS. Temporal regulation of shoot development in *Arabidopsis Thaliana* by miR156 and its target *SPL3*. *Development*. 2006;133(18):3539–47.
- Jung JH, Seo PJ, Kang SK, Park CM. miR172 signals are incorporated into the miR156 signaling pathway at the *SPL3/4/5* genes in *Arabidopsis* developmental transitions. *Plant Mol Biol*. 2011;76(1–2):35–45.
- Yao T, Park BS, Mao HZ, Seo JS, Ohama N, Li Y, et al. Regulation of flowering time by *SPL10*/MED25 module in *Arabidopsis*. *New Phytol*. 2019;224(1):493–504.
- Shao Y, Zhou HZ, Wu Y, Zhang H, Lin J, Jiang X, et al. *OsSPL3*, an SBP-domain protein, regulates crown root development in rice. *Plant Cell*. 2019;31(6):1257–75.
- Si L, Chen J, Huang X, Gong H, Luo J, Hou Q, et al. *OsSPL13* controls grain size in cultivated rice. *Nat Genet*. 2016;48(4):447–56.
- Jiao Y, Wang Y, Xue D, Wang J, Yan M, Liu G, et al. Regulation of *OsSPL14* by OsmiR156 defines ideal plant architecture in rice. *Nat Genet*. 2010;42(6):541–4.
- Wang J, Zhou L, Shi H, Chern M, Yu H, Yi H, He M, Yin J, Zhu X, Li Y, Li W, Liu J, Wang J, Chen X, Qing H, Wang Y, Liu G, Wang W, Li P, Wu X, Zhu L, Zhou JM, Ronald PC, Li S, Li J, Chen X. A single transcription factor promotes both yield and immunity in rice. *Science*. 2018;361(6406):1026–8.
- Klein J, Saedler H, Huijser P. A new family of DNA binding proteins includes putative transcriptional regulators of the *Antirrhinum majus* floral meristem identity gene SQUAMOSA. *Mol Gen Genet*. 1996;250(1):7–16.
- Hou H, Jia H, Yan Q, Wang X. Overexpression of a SBP-Box gene (*VpSBP16*) from Chinese wild vitis species in *Arabidopsis* improves salinity and drought stress tolerance. *Int J Mol Sci*. 2018;22(4):940.
- Li X, Shen F, Xu X, Zheng Q, Wang Y, Wu T, et al. An HD-ZIP transcription factor, *MxHB13*, integrates auxin-regulated and juvenility-determined control of adventitious rooting in *Malus xiaojinensis*. *Plant J*. 2021;107(6):1663–80.
- Zheng Y, Wang N, Zhang Z, Liu W, Xie W. Identification of flowering regulatory networks and hub genes expressed in the leaves of *Elymus sibiricus* L. using comparative transcriptome analysis. *Front Plant Sci*. 2022;13:877908.
- De Y, Yan W, Gao F, Mu H. Unraveling the signaling pathways of phytohormones underlying salt tolerance in *Elymus sibiricus*: a transcriptomic and metabolomic approach. *Genomics*. 2024;116(5):110893.
- Yan J, Li X, Wang L, Li D, Ji C, Yang Z, Chen L, Zhang C, You M, Yan L, Gou W, Lei X, Ji X, Li Y, Wu Q, Mao D, Chang D, Jia S, Li P, Zhang J, Xiong Y, Xiong Y, Han M, Chen Z, Cheng X, Tang J, Xie W, Liu W, Zheng H, Ma X, Yan X, Bai S. A high-continuity and annotated reference genome of allotetraploid Siberian Wildrye (*Elymus sibiricus* L., poaceae: Triticeae). *BioRxiv*. 2024;04(17):589894.
- Shen W, Liu B, Guo J, Yang Y, Li X, Chen J, et al. Chromosome-scale assembly of the wild cereal relative *Elymus sibiricus*. *Sci Data*. 2024;11(1):823.
- Chen CJ, Hao C, Yi Z, Thomas HR, Xia R. TBtools: an integrative toolkit developed for interactive analyses of big biological data. *Mol Plant*. 2020;13(8):1194–202.
- Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombauts S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in Silico analysis of promoter sequences. *Nucleic Acids Res*. 2002;30(1):325–7.
- He B, Gao S, Lu H, Yan J, Li C, Ma M, et al. Genome-wide analysis and molecular dissection of the *SPL* gene family in *Fraxinus mandshurica*. *BMC Plant Biol*. 2022;22(1):451.
- Corréa LG, Riaño-Pachón DM, Schrago CG, dos Santos RV, Mueller-Roeber B, Vincenz M. The role of bZIP transcription factors in green plant

- evolution: adaptive features emerging from four founder genes. *PLoS ONE*. 2008;3(8):e2944.
38. Lee J, Lee I. Regulation and function of SOC1, a flowering pathway integrator. *J Exp Bot*. 2010;61(9):2247–54.
 39. Werner S, Bartrina I, Schmülling T. Cytokinin regulates vegetative phase change in *Arabidopsis Thaliana* through the miR172/TOE1-TOE2 module. *Nat Commun*. 2021;12(1):5816.
 40. Xu M, Hu T, Zhao J, Park MY, Earley KW, Wu G, et al. Developmental functions of miR156-regulated SQUAMOSA promoter binding protein-like (*SPL*) genes in *Arabidopsis Thaliana*. *PLoS Genet*. 2016;12(8):e1006263.
 41. Lai T, Wang X, Ye B, Jin M, Chen W, Wang Y, et al. Molecular and functional characterization of the SBP-box transcription factor *SPL*-CNR in tomato fruit ripening and cell death. *J Exp Bot*. 2020;71(10):2995–3011.
 42. He A, Zhou H, Ma C, Bai Q, Yang H, Yao X, Wu W, Xue G, Ruan J. Genome-wide identification and expression analysis of the *SPL* gene family and its response to abiotic stress in barley (*Hordeum vulgare* L.). *BMC Genomics*. 2024; 9;25(1):846.
 43. Zhong H, Kong W, Gong Z, Fang X, Deng X, Liu C, et al. Evolutionary analyses reveal diverged patterns of SQUAMOSA promoter binding protein-like (*SPL*) gene family in *Oryza genus*. *Front Plant Sci*. 2019;8(10):565.
 44. Peng X, Wang Q, Zhao Y, Li X, Ma Q. Comparative genome analysis of the *SPL* gene family reveals novel evolutionary features in maize. *Genet Mol Biol*. 2019;42(2):380–94.
 45. Kong H, Landherr LL, Frohlich MW, Leebens-Mack J, Ma H, dePamphilis CW. Patterns of gene duplication in the plant SKP1 gene family in angiosperms: evidence for multiple mechanisms of rapid gene birth. *Plant J*. 2007;50(5):873–85.
 46. Waqas M, Azhar MT, Rana IA, Azeem F, Ali MA, Nawaz MA, et al. Genome-wide identification and expression analyses of WRKY transcription factor family members from chickpea (*Cicer arietinum* L.) reveal their role in abiotic stress-responses. *Genes Genomics*. 2019;41(4):467–81.
 47. Liu H, Lin M, Zhou D, Liu B, Li X, Wang H, et al. Characterization of the m⁶A gene family in Switchgrass and functional analysis of *PvALKBH10* during flowering. *Plant Physiol Biochem*. 2024;212:108781.
 48. Singh S, Praveen A, Bhadrecha P. Genome-wide identification and analysis of *SPL* gene family in chickpea (*Cicer arietinum* L.). *Protoplasma*. 2024;261(4):799–818.
 49. Jadhao KR, Kale SS, Chavan NS, Janjal PH. Genome-wide analysis of the *SPL* transcription factor family and its response to water stress in sunflower (*Helianthus annuus*). *Cell Stress Chaperones*. 2023;28(6):943–58.
 50. Xue G, Wu W, Fan Y, Ma C, Xiong R, Bai Q, et al. Genome-wide identification, evolution, and role of *SPL* gene family in beet (*Beta vulgaris* L.) under cold stress. *BMC Genomics*. 2024;25(1):101.
 51. Weits DA, Kunkowska AB, Kamps NCW, Portz KMS, Packbier NK, Nemec Venza Z, et al. An apical hypoxic niche sets the pace of shoot meristem activity. *Nature*. 2019;569(7758):714–7.
 52. Shukla V, Lombardi L, Iacopino S, Pencik A, Novak O, Perata P, et al. Endogenous hypoxia in lateral root primordia controls root architecture by antagonizing auxin signaling in *Arabidopsis*. *Mol Plant*. 2019;12(4):538–51.
 53. Pfeiffer A, Janocha D, Dong Y, Medzihradsky A, Schöne S, Daum G, Suzuki T, Forner J, Langenecker T, Rempel E, Schmid M, Wirtz M, Hell R, Lohmann JU. Integration of light and metabolic signals for stem cell activation at the shoot apical meristem. *Elife*. 2016;5:e17023.
 54. İlhan E, Büyük İ, İnal B. Transcriptome - scale characterization of salt responsive bean TCP transcription factors. *Gene*. 2018;5:642:64–73.
 55. Rouster J, Leah R, Mundy J, Cameron-Mills V. Identification of a methyl jasmonate-responsive region in the promoter of a Lipoxigenase 1 gene expressed in barley grain. *Plant J*. 1997;11(3):513–23.
 56. Zhu T, Liu Y, Ma L, Wang X, Zhang D, Han Y, et al. Genome-wide identification, phylogeny and expression analysis of the *SPL* gene family in wheat. *BMC Plant Biol*. 2020;20(1):420.
 57. Xie K, Wu C, Xiong L. Genomic organization, differential expression, and interaction of SQUAMOSA promoter-binding-like transcription factors and microRNA156 in rice. *Plant Physiol*. 2006;142(1):280–93.
 58. Zhang SD, Ling LZ, Yi TS. Evolution and divergence of SBP-box genes in land plants. *BMC Genomics*. 2015;16:787.
 59. Liu M, Sun W, Ma Z, Huang L, Wu Q, Tang Z, et al. Genome-wide identification of the *SPL* gene family in Tartary buckwheat (*Fagopyrum tataricum*) and expression analysis during fruit development stages. *BMC Plant Biol*. 2019;19(1):299.
 60. Zhu D, Geng X, Zeng F, Xu S, Peng J. Identification and expression analysis of the *SPL* gene family during flower bud differentiation in rhododendron molle. *Genes Genomics*. 2025;47(2):171–82.
 61. Dong F, Song J, Zhang H, Zhang J, Chen Y, Zhou X, et al. *TaSPL6B*, a member of the squamosa promoter binding protein-like family, regulates shoot branching and florescence in *Arabidopsis Thaliana*. *BMC Plant Biol*. 2024;24(1):708.
 62. Li Q, Yang Q, Dong S, Fu F, Xin Y, Kang H, et al. Transcription factors *CpSPL5* and *CpSPL8* negatively regulate salt tolerance in *Codonopsis pilosula* by inhibiting SOS pathway. *Plant J*. 2025;121(2):e17205.
 63. Sun X, Zhang L, Xu W, Zheng J, Yan M, Zhao M, Wang X, Yin Y. A comprehensive analysis of the peanut SQUAMOSA promoter binding protein-like gene family and how *AhSPL5* enhances salt tolerance in Transgenic *Arabidopsis*. *Plants (Basel)*. 2024;93(8):1057.
 64. Zhu J, Li Y, Zhang Y, Xia L, Hu W, Huang X, et al. Overexpression of *MiSPL3a* and *MiSPL3b* confers early flowering and stress tolerance in *Arabidopsis Thaliana*. *Int J Biol Macromol*. 2024;262(Pt 1):129913.
 65. Yu S, Galvão VC, Zhang YC, Horrer D, Zhang TQ, Hao YH, et al. Gibberellin regulates the *Arabidopsis* floral transition through miR156-targeted SQUAMOSA promoter binding-like transcription factors. *Plant Cell*. 2012;24(8):3320–32.

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